CHRISTEYE / SYLPH LABORATORY PROGRAM

EXPERIMENTAL DESIGN

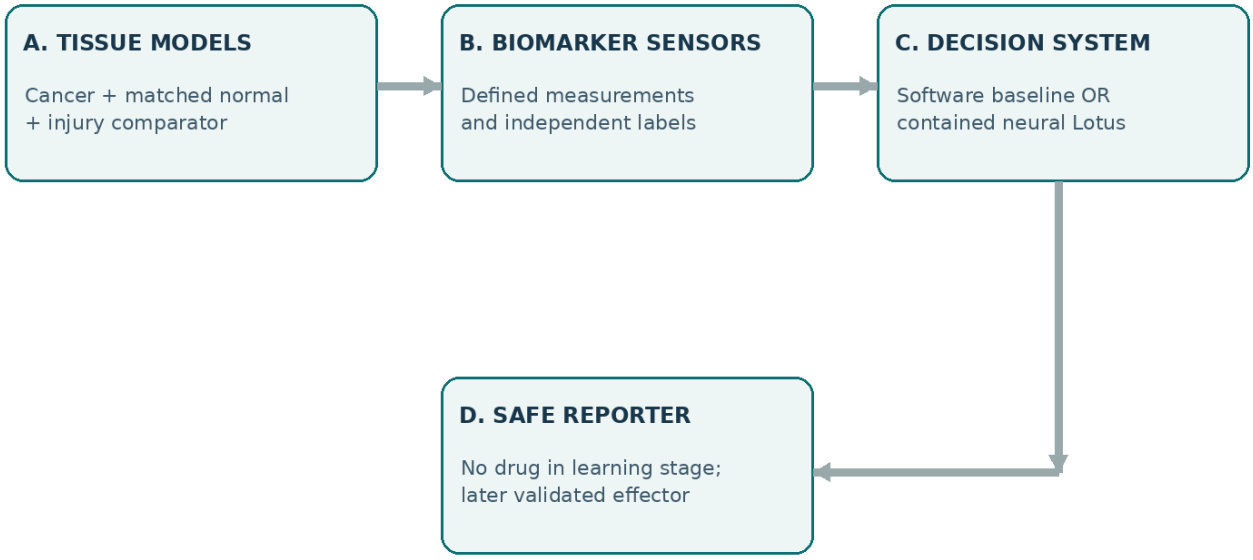
Living Periodic Table • Neural & Axon Training • Cancer-Cell Substrate Testing

Technical research specification | Version 1.1 | 2 November 2026

SCIENTIFIC SCOPE

This is a staged, nonclinical protocol design for qualified research laboratories—not a recipe for a patient vial, not a human dosing recommendation, and not proof of cancer prevention or cure. No cells or substrates described here should be swallowed, injected, implanted, or used outside approved laboratory oversight.

Prepared for We Develop Software LLC. Builds on the ChristEye Living Table & Energy Framework (v0.5), ChristEye Sylph Cancer Surveillance Concept (v0.2), and Sylph Research & Trials Plan (v1.0).



Independent stop rules and blinded outcome measurements apply to every arm.

PRIMARY TEST | Does closed-loop neural training improve detection and controlled response to cancer-associated signals beyond an ordinary software controller, without increasing tumor growth or harming normal cells?

01 Study question and claim boundaries

Sylph proposes a modular system in which biomarkers are detected by conventional sensors, a Lotus controller interprets those measurements, an independently governed safety system decides whether intervention is allowed, and a specialized cell-based or non-cell-based effector performs a separately validated task. The initial laboratory work is deliberately external to patients and separates computational learning from cancer-cell response.

Two hypotheses, not one

Hypothesis	Falsifiable primary comparison	Interpretation
H1 / recognition	Trained cultured-neuron controller vs ordinary software and untrained neurons on held-out normal / injury / malignant signal patterns.	Tests incremental classification, calibration, stability, and signal retention; does not imply cancer treatment.
H2 / therapeutic regulation	Same independently validated anticancer effector governed by software vs trained neural controller, under identical safety rules.	Tests tumor-selective benefit and normal-tissue harm; start only after H1 passes.
H3 / danger sentinel	Neuron-associated secretions or signals compared with no-neuron reference under separate oversight.	Tests whether neural context unintentionally accelerates cancer proliferation.

A negative result is valuable: if software performs as well as or better than neurons, the living-controller claim is not supported. If neuron-associated signals increase proliferation or invasion, the direct tissue-contact concept is rejected or redesigned.

02 How to read the Living Periodic Table

The Living Table is your functional taxonomy across levels: chemical matter, molecular components, organelles, differentiated cells, tissue structures, information, and coordinated purpose. It is not a chemical element list to be mixed into a vial. Biological materials require appropriate identity, context, concentration, and compatibility; more ingredients do not imply greater effectiveness.

ChristEye category	System meaning	Testable analogue
Soul	Integrated identity / continuity of the therapeutic system	Reproducible function, integrity, defined interfaces and controlled persistence
Lotus	Action, growth, coordination and trained response	Sensing-to-output controller, task performance, plasticity
Marrow	DNA, lineage and inherited structure	Cell identity, genomic stability, verified differentiation
Roots	Connectivity, history and retained learning	Synapse/axon structure, memory scores, longitudinal baseline
Winds	Signal flow and activity	Electrode measurements, ions, cytokines, metabolites, electrical output
Heart	Needs, priorities, biological homeostasis	Risk thresholds, viability, metabolic stress, independent stop conditions
Spirit / knowledge	Initiating inputs and learned relationships in the conceptual model	Defined experimental stimuli; encoded data and predictive associations
Understanding	Robust, appropriate behavior over time	Generalization, low false-alarm rates, retention, no net harm

INTERPRETATION RULE

Soul, Spirit and the other ChristEye names preserve the philosophical architecture. They do not substitute for biological identification, measurements, chemistry or evidence of cancer treatment.

03 Candidate living components and functional measurements

Level A: ordinary chemical elements, ions and nutrients

Group	Named components	Role and measurement—not a patient dose
Major biogenic elements	C, H, O, N, P, S	Cell structures, proteins, nucleic acids and metabolism; track through certified culture-media composition and mass balance.
Neuronal ions	Na^+, K^+, Ca^{2+}, Mg^{2+}, Cl^-	Electrical excitability, calcium signaling, enzymatic function and osmotic stability; assess against validated media standards.
Trace elements	Fe, Zn, Cu, Se, Mn	Enzyme/cofactor roles; excessive amounts may be toxic. Qualify rather than add independently.
Core nutrient classes	Glucose, amino acids, lipids, vitamins, oxygen	Required for cell survival and ATP production; monitor supplies and media compatibility.
Structural environment	Defined extracellular matrix, hydrogel or transwell substrate	Physical support and compartmentalization; measure compatibility, diffusion and batch variability.

Level B: molecules and organelles

Code	Element	Purpose and evaluation
DNA / RNA	Genomic DNA, mRNA and other RNAs	Information and gene expression; assay specific biomarkers, not mRNA as a universally dangerous material.
Rb	Ribosome	Protein production; product identity and expression fidelity.
Mt	Mitochondria	ATP metabolism; viability, energy demand, stress burden.
Sy	Synapses	Plastic connectivity; strength, response reliability, stability.
Ve	Vesicles and receptors	Chemical signaling; confirm intended targets, release and off-target activity.
Ax	Axons and cytoskeleton	Conduction and transport; propagation delay, failure, length/morphology, unwanted outgrowth.
Ch	Ion channels / transporters	Excitability and homeostasis; evoked signal

Code	Element	Purpose and evaluation
		fidelity and recovery.

Level C: cells, progenitors and cancer models

Code / candidate	First-generation role	Why not place in a universal vial
Nr / mature neurons	Isolated, electrical processing and learning research	Do not circulate as a stable network; can stimulate gliomas.
As / astrocytes	Support neural culture and control ion/metabolic environment	Secreted factors could influence tumor biology.
Ol / oligodendrocyte lineage	Later mechanism study of myelin and transmission	CNS-specific; does not confer cancer recognition.
Sc / Schwann cells	Later peripheral-axon comparator	Unnecessary for first tumor study; may alter innervation.
Mi / microglia	Research neuroinflammation and clearance signals	Context-dependent inflammatory/tumor effects.
Ns / neural stem cells	Model neural differentiation and targeted delivery literature	Uncontrolled proliferation/differentiation is unacceptable.
Nc / neural crest progenitors	Research migration and lineage choice in separated workstream	Developmental migratory cells with variable fates.
iPSC-derived cells	Reproducible source for well-characterized neural models	Residual pluripotency and genomic instability hazards.
MSC / stromal cells	Explore repair signaling and extracellular factors	Can also promote malignant growth; not default therapy.
HSC / hematopoietic lineage	Reference to established transplantation and immune lineage	Not a neural cancer repair ingredient.
T cells / NK cells	Specialist-defined anticancer-effector comparator	Require cancer antigen/indication and independent testing.
Tumor + nonmalignant tissue	Target and specificity reference models	Human cancer cells require approved laboratory containment.

Selection for prototype v1: cancer models + compatible normal models, a standard software comparator, one qualified cultured-neuron source, optional astrocyte support, transduction electrodes, and a harmless reporter. All other cell populations remain optional mechanism studies unless a predefined hypothesis justifies their inclusion.

04 Define the laboratory system—NOT a patient vial

Three physically separate compartments

1. Tissue compartment: authenticated cancer model and separate compatible nonmalignant or injury models; standard tissue-culture conditions established by the host laboratory.
2. Signal/learning compartment: neurons on a multielectrode array or equivalent readout in a dedicated culture system; astrocytes only if needed for stable function, with separate controls.
3. Controller/reporter compartment: instrument software encodes validated sensor data as inputs, reads the neural response, and produces a harmless reporter or simulated treatment decision. Independent rule-based safety gate cannot be bypassed by the neural controller.

Keep the living neuronal network physically isolated from the cancer model in the first pass. Biological communication is not the default: measured sensor data are carried through instrumentation. A later secretome or contact study would need a separate ethics, biosafety and tumor-promotion review.

Candidate supplies and quantity status

Research component	Illustrative planning amount	Final amount set by
Tumor models	Two distinct cancer models (one authenticated reference line; one patient-derived model if ethical access allows).	Oncology laboratory / material availability
Nonmalignant models	Two compatible normal or injury reference models, matched by relevant tissue biology when feasible.	Pathologist / tissue-model specialist
Neural controller preparation	One well-characterized neuronal source; compare cultured neurons alone vs supported only after pilot QC.	Neural-culture specialist; no arbitrary cell count
Sensor channels	Minimum three predefined signal categories: baseline, controlled injury, and malignancy-associated signatures.	Oncology biomarker team
Hydrogel / support substrate	One qualified baseline matrix and one material-only sham; do not invent a nutritional composition.	Materials / GMP/quality specialist
Neural safety monitoring	All experimental conditions: independent electrical limits, viability monitoring and emergency stop.	Bioelectronic engineer / lab safety officer
Therapeutic cells or agents	None in early learning experiment. Later: one separately characterized comparator under specialist supervision.	Licensed oncology research facility
Patient vial	Zero: no in-human, oral or blood-borne formulation is specified or authorized.	Only after extensive evidence and regulatory review

WHY EXACT CELL DOSE IS TBD

The cell count, per-well density, substrate formulation, concentration and time of exposure depend on the authenticated cell model, assay platform, viability and approved SOP. Giving a universal number now would be an invented and potentially misleading treatment or culture recipe. This plan defines measurable dose-selection criteria and records them as controlled, versioned laboratory parameters.

05 Quantitative pilot design and allocation

The following counts are an illustrative resource allocation—not a power calculation or a wet-lab preparation recipe. A statistician must confirm independent biological sample numbers after pilot variance is known. Different tissue models may require distinct media, vessels or laboratories; the units below represent analysis slots, not one physically mixed plate.

Item	Planning value	Calculation / note
Disease contexts	4 model types	2 tumor + 2 normal/injury models
Conditions	6 conceptual arms	Define below; some arms are model-specific
Technical repeat units	3 per model-condition	Reduce measurement noise; not independent donors

Item	Planning value	Calculation / note
One run assay slots	72	4 × 6 × 3 = 72 conditions measured
Independent run batches	3 exploratory	3 days or independent culture batches, if qualified
Total planned assay units	216	72 × 3, excluding QC and optional reserves
QC and controls	~20–30% additional capacity	Blanks, calibration and media-specific checks; allocate as needed
Biological independence	Not established by 216 wells	Different days ≠ different patient donors; assess clustering

Six-arm pilot comparison

Arm	Model input	What it controls / expected result
A / native state	Tissue model with no new Sylph interface.	Baseline proliferation, stress and survival.
B / substrate sham	Same model with qualified matrix/interface material alone.	Detect toxicity or growth effects of substrate.
C / conventional logic	Valid sensors + software classifier + harmless report.	Baseline accuracy and latency.
D / untrained Lotus	Sensors + cultured neurons, no informative learning feedback + harmless report.	Controls for mere presence of living neural activity.
E / trained Lotus	Same sensors + trained neurons + identical reporter and safety gate.	Tests whether contingent feedback improves decision performance.
F / independent assay control	Qualified assay positive/negative benchmark selected by oncology team.	Checks assay sensitivity; not proof of Sylph efficacy.

Pilot phases A–E have NO cancer-killing component. Tumor-cell readouts during them measure whether the system is non-tumor-promoting, not whether Sylph heals cancer. Therapeutic effect testing is a subsequent work package with a separately authorized comparator.

06 Step-by-step laboratory study workflow

WP0 | lock models, safety, questions

4. Select ONE indication for initial validation; glioblastoma offers a relevant but unusually high-risk neuron–tumor interaction model. Do not claim universal cancer prevention from this model.
5. Have institutional researchers approve cell provenance, biosafety procedures, neural material ethics and independent stop conditions. Assign sample IDs, blind readout analysts, and preregister the primary comparison and adverse-effect criteria.
6. Validate biological labels for normal, tissue injury and malignant states from orthogonal assays. Avoid equating inflammation alone with cancer. [S8]

WP1 | establish baseline cell-model readouts

7. Obtain authenticated, ethically sourced tumor and nonmalignant reference models under existing institutional protocols; use appropriate control media and verify model identity.
8. Characterize viability, cell proliferation and tissue injury in each model before attaching sensors or introducing neural signals.

9. Define simple synthetic or archived biomarker input patterns to train the neural controller without exposing cultures to an active clinical therapy.

WP2 | establish neural viability and stable axon signaling

10. Assess neuronal identity and network viability; capture spontaneous activity, evoked response, and stable signal pathways using a calibrated electrical instrument.
11. Quantify axonal propagation delay, transmission success, morphology, and changes over time. Identify any evidence of excessive growth or increasing metabolic stress.
12. Use astrocyte-supported and unsupported cultures only as an explicitly controlled comparison. Consider oligodendrocytes and Schwann cells as later, lineage-appropriate substudies.

WP3 | closed-loop neural training

13. Encode pre-labeled signal patterns into conservative, instrument-approved stimulation/representation channels. An independent decoder converts network recordings into a defined class prediction.
14. Compare contingent feedback with yoked or replayed feedback, stimulus-only, untrained neurons, and non-living software classification. Hold out entire culture batches and input patterns.
15. Assess whether training improves discrimination AND calibration, decreases errors, retains learning after a delay, and remains within viability and electrical safety limits.

WP4 | tumor specificity and harm monitoring

16. Pass blinded measurements of normal, injured and cancerous models through the decision systems. Keep tissue readout teams unaware of the decision-arm label.
17. Quantify correct malignant-state alerts, false alarms on injured nonmalignant tissue, missed cancer alerts, latency, drift and biological stress.
18. Analyze whether any material interface or neural-conditioned secretion increases tumor proliferation/invasion; stop or redesign if tumor-promoting evidence appears. [S6–S7]

WP5 | only after gates: functional treatment comparison

19. An oncology collaborator identifies a separately validated, indication-appropriate therapeutic effector or drug benchmark and its established assay context.
20. Compare fixed treatment, software-gated treatment and neuronal-controller-gated treatment with the SAME therapeutic effector and independent safety rules. Include healthy-tissue toxicity assessment.
21. If there is no selective incremental benefit, stop the therapeutic-controller branch. Do not infer effectiveness from tumor exposure, electrical response or reporter activation alone.

WP6 | repeat, independently reproduce, and decide

22. Repeat with an independent neuronal batch, another tumor model and separate blinded analysis. Preserve raw data, protocols, failures and adverse results.
23. Only following reproducible nonclinical safety and target-specific benefit, consult regulators and clinical research experts about further preclinical work; no human dosing is proposed in this document.

07 Training program: neurons, axons and support cells

7.1 Distinguish what is actually trained

Biological material	What can change	Training metric / constraint
Mature neurons	Response patterns and network connectivity under feedback.	Held-out prediction metrics, response repeatability, survival.
Axons	Conduction and network function can change through development and activity.	Propagation fidelity, delay, unwanted sprouting and energy demand.

Biological material	What can change	Training metric / constraint
Synapses	Effective connection weights can strengthen or weaken.	Evoked network responses, retention and calibration.
Astrocytes	Support environment, metabolism and neuronal signaling.	Ion/metabolic stability, viability; check secreted factors.
Oligodendrocytes / Schwann	Myelination and axon support in applicable systems.	Transmission effects and lineage fidelity; later sub-study only.
Neural stem cells	Differentiate along regulated lineages, not learn a cancer diagnosis like a mature neural network.	Lineage identity, residual stemness, genetic stability and proliferation.
Neural crest progenitors	Differentiate and migrate depending on developmental cues.	Migration and lineage results; not default cancer sensor.
Microglia / macrophages	Respond to immune/inflammatory cues.	Context specificity and risk of tissue injury.

7.2 Four progressive training tasks

Task	Design principle	Pass/fail observable
T1 / signal identity	Distinguish labeled neutral patterns unrelated to cancer.	Improvement over untrained and chance, independent batches.
T2 / three-state classification	Differentiate baseline vs injury-like vs malignant-associated proxy signals.	Sensitivity, specificity, macro-F1, AUROC for prespecified labels.
T3 / temporal understanding	Respond appropriately when the same signal repeats or resolves.	Fewer false alerts with reversible injury; stable recall and recovery.
T4 / safe decision timing	Propose a harmless output only when prespecified criteria are met.	Appropriate no-action decisions, latency, shutdown compliance.

7.3 Axon training scorecard

Metric	Definition	Illustrative decision expectation
Propagation success	Fraction of evoked signals reaching a prespecified downstream readout.	Must not deteriorate relative to baseline; numerical range learned in pilot.
Latency and jitter	Time to signal arrival and variability.	Stable within validated instrument and biological ranges.
Plasticity / retention	Change in task-relevant mapping retained on unseen signals.	Better than untrained/replayed feedback; no overfitting.
Network drift	Change in prediction when controls and inputs are held stable.	Within a preregistered limit; drift triggers retraining review.
Energy per correct decision	Measured energy use over correctly classified trials.	Compare with software; energy gain is not assumed.
Viability / unintended growth	Morphology, death, proliferation, outgrowth beyond assigned region.	Any critical safety threshold failure halts progression.

Training and routine sensing cannot be used interchangeably: a culture that reacts to a conditioned input has not automatically learned the biology of cancer. All learning claims require held-out controls and the same labels available to the software comparator. [S4–S5]

08 Stem-cell differentiation and controlled therapeutic roles

Stage-by-stage cell selection

Cell type	Proposed experiment	Gate and rationale
iPSC-derived neuronal progenitors	Generate a stable, typed research neural culture from a certified source.	Confirm identity and lack of unwanted undifferentiated cells; not used clinically.
Neural stem cells	Separate specialized study of tumor-directed delivery in glioma models.	Independent oversight for migration, persistence, unintended growth.
Neural crest cells	Optional migration/lineage reference in a different experimental series.	Do not combine with initial oncology effectors.
Mesenchymal stromal cells	Optional secretome control to test wound-like vs tumor-supporting signals.	Reject if proliferation/invasion increases.
Hematopoietic stem cell lineage	Historical/clinical comparator for transplantation, not trial ingredient.	Exclude from mixed-vial formulation.
T/NK or specialist therapeutic cells	Later qualified, validated anticancer effectors only.	Must be independently assessed for cancer selectivity, safety and authorization.

Three different controls—not interchangeable

- INTRODUCTION: Only qualified researchers decide when and where cells enter an experimental compartment. This is not equivalent to a cell spontaneously appearing on demand.
- ACTIVATION: Validated conditional gene expression or cell activity can be investigated with harmless reporters first; an independent controller should enforce decision thresholds.
- PERSISTENCE: Cell proliferation, migration and viability require separate containment and, for any eventual engineered effector, independently verified elimination safeguards. Never assume 100% shutdown or that all stem cells are beneficial.

09 Cancerous-cell substrate experiment: defined readouts

Model choice and outcome hierarchy

For one initial indication, work with a specialist laboratory to use an authenticated malignant model and tissue-matched nonmalignant comparator. Patient-derived organoids may be added later because tumor heterogeneity changes how therapies behave. Models are not interchangeable across media or exposure conditions. [S9]

Endpoint class	Examples of measurement	Why it matters
Cancer viability and burden	Quantitative viable-cell counts; tumor size or growth trend.	Primary efficacy proxy once a validated anticancer effector exists.
Cancer proliferation and death	Proliferation labeling, apoptosis markers, live imaging.	Distinguish growth inhibition from nonspecific toxicity.
Healthy tissue preservation	Normal-cell viability, morphology, differentiation and injury.	Avoid trading cancer harm for comparable healthy-tissue harm.
Neural control accuracy	Confusion matrix, AUROC, F1, precision-recall, calibration.	Evaluate claimed cognition independent of cancer killing.
Network physiology	Electrode activity, axon reliability, oxygen/ATP stress.	Prevent loss of function or overexcitation.
Off-target / tumor promotion	Glioma growth, invasion and changes after neural-conditioned contact.	Neurons may increase glioma progression; must test directly.

Endpoint class	Examples of measurement	Why it matters
Persistence and shutdown	Reporter off-state, viable-cell persistence, stop response.	Safety and reversibility; release claims require independent checks.

Illustrative measurements and timing

A qualified facility may predefine a baseline time point, early response window and later follow-up (for example, T0, T1 and T2), matched to the growth kinetics of the chosen models. The document intentionally does not prescribe tumor inoculation, stem-cell densities, culture recipes, electrical stimulation amplitudes, transfection methods or biologic dosing; these are platform- and safety-dependent controlled SOPs.

10 Analysis plan, targets and falsification

Primary comparison and statistics

For classification, prespecify macro-F1 and patient/batch-held-out discrimination as primary performance outcomes; report class-specific sensitivity, specificity, AUROC, precision-recall, confusion matrices, calibration error and false negatives. Test whether the neural controller adds incremental benefit over a non-living software controller trained on the exact same data. Do not count 3 technical wells as 3 independent patient samples.

For therapeutic work, prespecify tumor burden/growth response and nonmalignant cell viability as co-primary biological outcomes, use blinded readings and confidence intervals, and estimate variability before power calculations. Adjust for donor and batch clustering, multiplicity and between-model heterogeneity.

Illustrative development targets (NOT evidence of efficacy)

Candidate decision target	Provisional value	Caution
Classification improvement vs software	≥ 0.05 absolute macro-F1 gain in held-out evaluation	Aspirational project gate, not clinical diagnostic qualification.
Repeatability	Same directional effect in ≥3 independent culture batches	Does not equal validation in 3 patients or centers.
No tumor promotion	No repeatable increase vs comparable sham in growth or invasion	Requires adequately sensitive assays and independent replication.
Therapeutic selectivity	Meaningful tumor response with smaller healthy-tissue harm vs comparator	Effect threshold and confidence intervals chosen by oncologist and statistician.
Shutdown	Consistent stopping behavior during a deliberate, safe simulated-stop assessment	No laboratory stop control guarantees complete elimination in vivo.

Failure must be published internally

- If cultured neurons do not outperform software on held-out patterns: pivot to a software-only controller; neural contribution is unproven.
- If neuronal-conditioned signals increase malignant growth: discontinue neural–tumor contact concepts and consider the controller-only configuration.
- If the effector is not tumor-selective: no treatment claim and no progression to complex integration.
- If a cell population cannot be contained or reliably characterized: remove it from the design.

11 Safety, ethics, and product boundary conditions

Risk	Required mitigation
Tumor stimulation from neural signals	Maintain compartments separated in early tests; assess neuro–glioma signaling risk [S6–S7].
Tumorigenicity from stem/progenitor cells	Cell identity, genomic stability, residual pluripotency assays, persistence studies before any translational proposal [S10].
Infection / contamination	Institutional biosafety, qualified materials, donor screening and culture quality documentation [S1].
False-negative cancer detection	Report miss rates with uncertainty; no replacement of standard clinical diagnostics.
False-positive treatment activation	Independent safety rules, harmless early reporters, specialist review of clinical outputs.
Neural culture instability	Monitor survival, electrical drift and stress; no autonomy over dosing.
Oral or bloodstream deployment	Not a viable immediate path for a network of human neurons; not studied in this work.
Human studies	Not authorized. Require robust nonclinical evidence and FDA IND/ethics review before any clinical trial. [S1, S10]

EXPLICIT PROHIBITION FOR THE PROPOSAL

Do not attempt to swallow, inject, transfuse or implant cultured neurons, stem cells, scaffolds or homemade biologic preparations. A refrigerated vial and expiration date do not make a live-cell treatment safe, sterile or clinically effective.

12 Trial pathway after laboratory evidence

Gate	Required evidence	Decision
G0 / organizational	Qualified PI, facility, protocol approvals and reliable cell models.	Begin data/model validation only.
G1 / material safety	Identity, contamination QC, stable nonmalignant assays.	Begin cell-free and neural readouts.
G2 / neural benefit	Retained, held-out improvement over software; acceptable reliability.	Continue; otherwise software-only.
G3 / harm protection	No reproducible tumor promotion, preserved healthy-cell safety.	Continue; otherwise stop.
G4 / effector specificity	Independent oncologic benefit with bounded toxicity.	Only then integrate gating.
G5 / integrated reproducibility	Validated independent lab and safety controls.	Evaluate more advanced preclinical studies.
G6 / regulatory	CMC, toxicology, tumorigenicity, biodistribution and FDA guidance review.	Conditional IND planning; not automatic approval.
G7 / clinical if approved	Clinical investigators design first-in-human dose, safety and monitoring.	Human trial only with required authorization.

Regulatory details change by product type. FDA guidance on cellular/gene-therapy development, tumorigenicity and preclinical work must be interpreted by qualified specialists [S1, S10]. The mere availability of neural stem-cell glioma trials does not validate Sylph [S3].

13 Project timeline, team and handoff worksheets

Work period	Planning milestone	Deliverable
Months 0–2 / design	Choose indication, secure qualified collaborators, review biosafety and ethics.	Approved project charter; documented sourced materials.
Months 2–5 / models	Establish sensor signals, model integrity, software baseline.	Model/QC data; clinical label definitions.
Months 4–9 / neural	Characterize axonal functions and train basic recognition tasks.	Closed-loop logs, blinded error rates, viability data.
Months 8–12 / comparisons	Tissue specificity and neuron-associated tumor-promotion sentinel.	Independent assay results, error bars, risk register.
Months 12+ / contingent	Only if all gates pass: validated effector comparator, outside replication.	Go/no-go committee report and regulatory discussion plan.

Controlled parameters to be filled in by the experimental PI

Parameter	Record value	Required justification
Chosen tumor model, source and passage range	TBD	Identity, representative indication, contamination screening
Nonmalignant comparator source	TBD	Tissue match and clinical relevance
Neuronal culture composition and density	TBD	Validated neural platform SOP and viability
ECM/substrate lot, composition and volume	TBD	Compatibility and absence of tumor promotion
Electrode layout and allowed electrical envelope	TBD	Instrument calibration and cell safety
Training schedule and stopping rules	TBD	Pre-registered learning and drift plan
Sensor labels and assay concentrations	TBD	Biomarker validation, performance and safety
Effector selection and exposure range	Not used in Stage 1	Independent established oncologic research rationale
Sample size and statistical power	Pilot n shown above	Variance, unit of independence, desired effect

Minimum collaborators

A cancer biologist or oncologist; a neural electrophysiologist; a stem-cell/developmental biologist; a bioelectronic systems engineer; an assay specialist; a biostatistician; biosafety/ethics officers; and a regulatory CMC specialist for any eventual cell product. The independent analysis team should review both benefits and adverse signals.

14 Claims, non-claims and results template

What can be claimed after Stage 1, if supported

- A well-defined neural culture achieved or failed a specified, reproducible learning task using defined cancer-associated signal proxies.

- The living controller demonstrated or did not demonstrate incremental benefit beyond conventional software.
- Material and neural components did or did not show measurable adverse effects on the selected cancer and nonmalignant models.

What cannot be claimed from Stage 1

- Cancer cure, patient protection, prevention of all cancers, personalized systemic neural surveillance, body-wide healing, or safe daily vial administration.
- That neural axons can travel through blood to repair cells; that stem cells appear only when needed; that a trained network performs perfectly.

Results-report skeleton

Field	What to enter
Model identity and provenance	Authenticated cancer model, comparator, permissions, biological replicate count.
Pre-registered hypothesis	Primary endpoint and independent comparators.
Learning evidence	Training curve, held-out calibration, confusion matrices, software comparison.
Biological outcomes	Tumor growth/death, normal tissue injury, secretome hazard, viability.
Reliability	Culture batches, axon signaling, drift, instrument failures and QC exclusions.
Interpretation	Supported / unsupported / inconclusive; limitations; independent replication need.

15 Evidence basis and research sources

User-defined architecture comes from the ChristEye Living Table & Energy Framework v0.5, ChristEye Sylph Cancer Surveillance Concept v0.2, and ChristEye Sylph Research & Trials Plan v1.0. The sources below support specific mechanisms and regulatory constraints, not the efficacy of the complete Sylph system.

[S1] FDA, Frequently Asked Questions—Developing Potential Cellular and Gene Therapy Products, final guidance (Aug 2026).
https://www.fda.gov/regulatory-information/search-fda-guidance-documents/frequently-asked-questions-developing-potential-cellular-and-gene-therapy-products

[S2] NCI, CAR T Cells: Engineering Immune Cells to Treat Cancer (2025).
https://www.cancer.gov/about-cancer/treatment/research/car-t-cells

[S3] NCI, Neural Stem Cell Therapy for Recurrent High-Grade Gliomas (NCT05139056).
https://www.cancer.gov/research/participate/clinical-trials-search/v?id=NCI-2022-10170

[S4] Kagan et al., In vitro neurons learn and exhibit sentience when embodied in a simulated game-world, Neuron (2022).
https://pmc.ncbi.nlm.nih.gov/articles/PMC9747182/

[S5] Cai et al., Brain organoid reservoir computing for artificial intelligence, Nature Electronics (2023).
https://www.nature.com/articles/s41928-023-01069-w

[S6] Venkatesh et al., Electrical and synaptic integration of glioma into neural circuits, Nature (2019).
https://www.nature.com/articles/s41586-019-1563-y

[S7] Glioma synapses recruit mechanisms of adaptive plasticity, Nature (2023).
https://www.nature.com/articles/s41586-023-06678-1

[S8] National Cancer Institute, Chronic Inflammation and Cancer Risk.
https://www.cancer.gov/about-cancer/causes-prevention/risk/chronic-inflammation

[S9] Patient-derived tumor organoids mimic treatment-induced DNA damage response in glioblastoma, iScience (2024).
https://www.sciencedirect.com/science/article/pii/S2589004224018297

[S10] FDA, Preclinical Assessment of Investigational Cellular and Gene Therapy Products (2013) and FAQ (2026) for current context.
https://www.fda.gov/regulatory-information/search-fda-guidance-documents/preclinical-assessment-investigational-cellular-and-gene-therapy-products

[S11] FDA, Important Patient and Consumer Information About Regenerative Medicine Therapies.
https://www.fda.gov/vaccines-blood-biologics/consumers-biologics/important-patient-and-consumer-information-about-regenerative-medicine-therapies

[S12] NIH NINDS, Life and Death of a Neuron (2025 booklet).
https://www.ninds.nih.gov/sites/default/files/2025-05/NINDS_Life_and_Death_of_a_Neuron_Booklet_073024_B_FOR_ONLINE_0.pdf

16 Standardized experimental record — quantity and element ledger

This register makes the requested “amounts” explicit without presenting invented biological concentrations. Every numeric cell value must come from a qualified institutional SOP, an authenticated supplier certificate, or a preapproved method-development study. Data are retained for each model, culture batch and time point.

Field / unit	Example entry style	How the laboratory sets it
Cell identity / line or donor ID	ID + assay authentication report	Cell bank and PI
Viable cell count / cells and cells/mL	Measured; no default amount	Platform-specific viable counting QC
Material dose / mg, µg, µL, mL	Measured, lot recorded, range justified	Biocompatibility and matrix SOP
Nutrients / mmol/L or mg/L	Use certified medium specification	Media provider and cell physiology
Ions / mmol/L	Na+, K+, Ca2+, Cl−, Mg2+ measured or specified	Neural medium and instrument SOP
Exposure duration / h or day	T0, T1, T2 chosen by growth kinetics	Oncology model pilot
Neural stimulation / instrument units	Prevalidated amplitude and waveform	MEA safety envelope; independent shutoff
Output and survival / %, counts	Blindly measured, baseline normalized	Pre-registered assay analysis
Reporter / relative fluorescence etc.	Calibrated readout per condition	Assay quality control

Investigator signoff fields: tumor-model PI: _____ neural-culture PI: _____ biosafety review: _____
biostatistics: _____ protocol version/date: _____

STATUS OF ALL NUMERICAL VALUES

The 4 × 6 × 3 × 3 study allocation and development targets are illustrative assumptions for research budgeting, NOT validated sample-size requirements, drug doses, cell concentrations, nutrient recipes or measured ChristEye performance.

END OF DOCUMENT | ChristEye Sylph Experimental Design v1.1